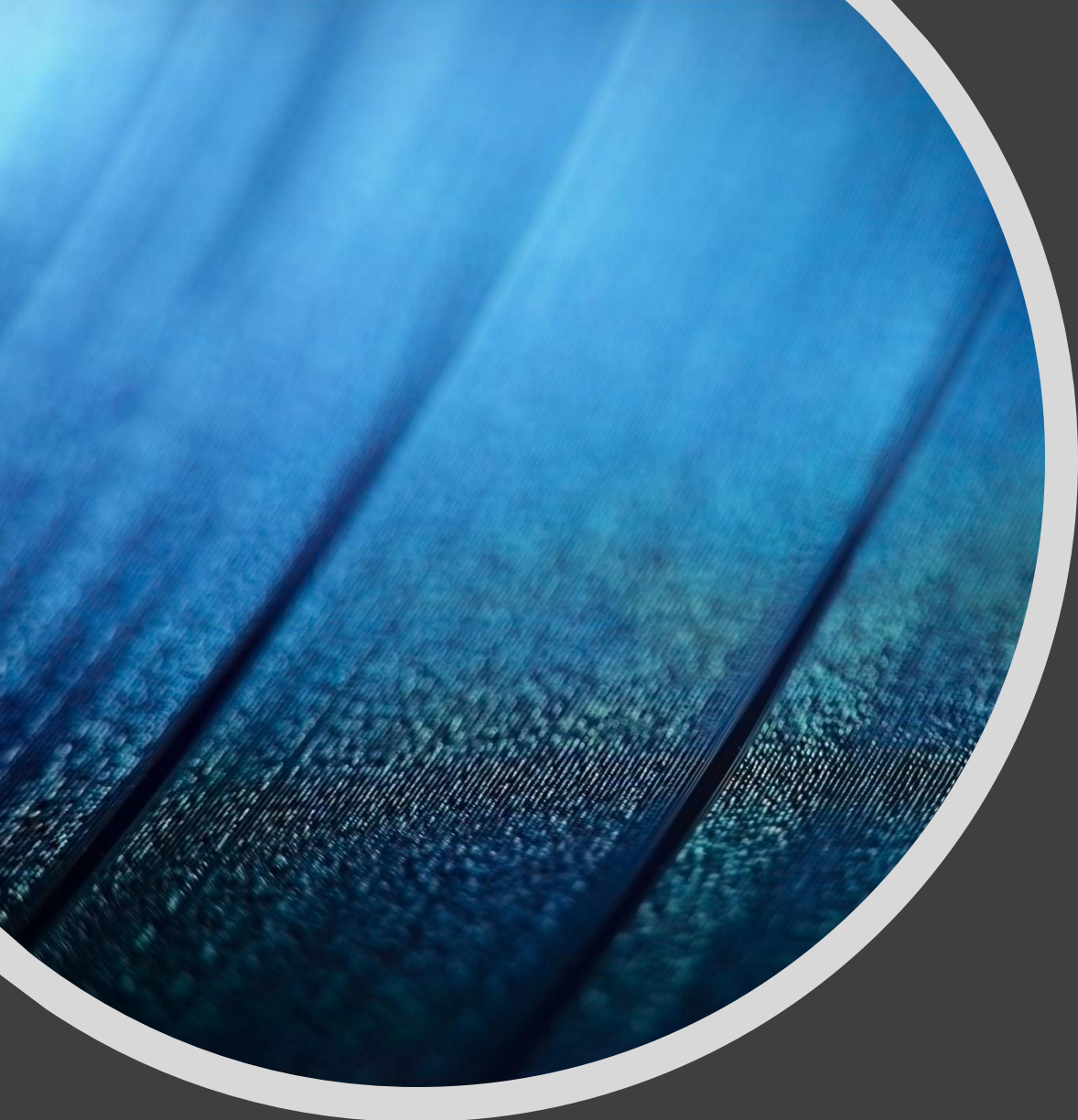


AI For the Better



2022 // CloudBrew// Dennie Declercq



Content

Intro

Accessibility & Disabilities

Microsoft Cognitive & Applied AI Services for the Better

Unsolved Real Life Problems to Fix

What is done before?

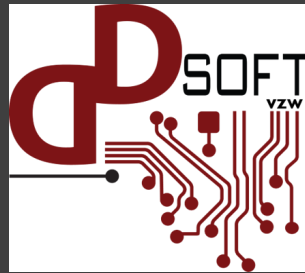
Inspiration for your next hack

Outro



About Me

- Dennie Declercq
- President/ Developer DDSOFT
- Microsoft MVP
- @DennieDeclercq
- #DreamingIsBelieving



Accessibility & Disabilities

The Disability Spectrum

- Touch

- One arm
- Arm injury
- New parent

- See

- Blind
- Low vision
- Being distracted

- Hear

- Deaf
- Hard of hearing
- Ear infection

- Speak

- Non-verbal
- Heavy accent
- Mouth infection

- Cognition

- Learning disability
- Down syndrome
- Memory difficulties

- Mental Health

- Neurodiversity
- Anxiety
- Confused/ stressed (bad news)

Learning Disability

Intellectual Functioning

Also called intelligence

Mental capacity

IQ age between 2 -8yo

Adaptive Behavior

Conceptual Skills

- Language/literacy
- Money
- Time/numbers

Social Skills

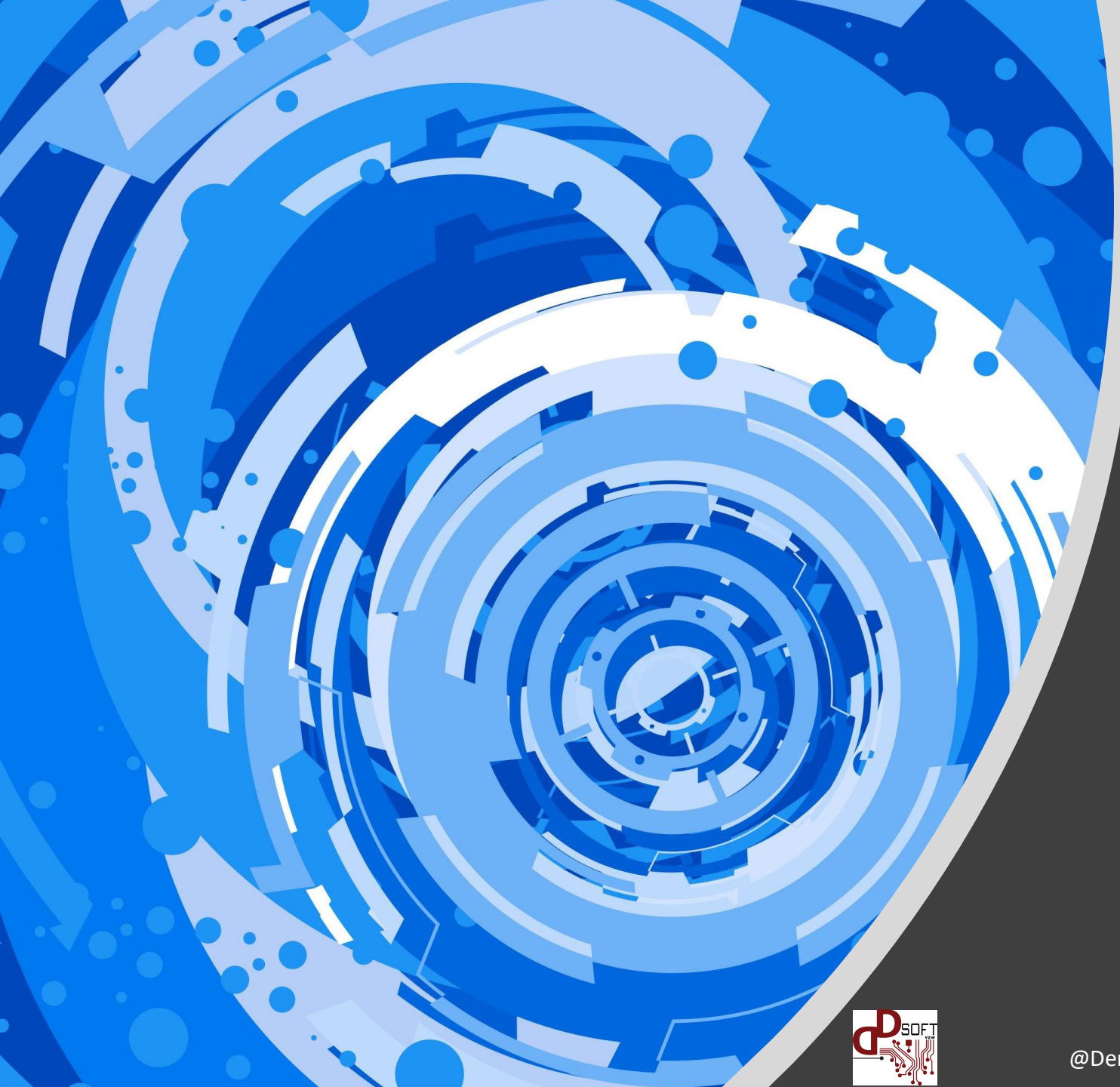
- Social responsibility
- Self-esteem

Practical skills

- Activities daily living
- Personal care / healthcare

Mental Health & Autism

- Stress and Anxiety
- Need for confirmations
- Need for structure
- Need for easy to understand information



Microsoft Cognitive & Applied Services for the Better





What are Microsoft Cognitive Services?

- Microsoft Developer Focussed AI platform
- Easy to use API's
- Multiple language technologies (C#, JAVA, Python...)
- REST compatibility
- Supported SDK's
- No AI or Mathematical knowledge needed
- Build in the Azure Suite
- Pay as you Go/ Pay by use
- Serverless
- Affordable



Immersive Reader

- Read aloud text
 - Read aloud websites
 - Linking pictographs to websites
 - Text appearance options
 - Colorize verbs/ nouns
- Make written text accessible
 - People with learning disabilities
 - People with Dyslexia
 - Elderly
 - Students



Computer Vision

- Analyze Images
- Label Images
- Categorize Images
- Measures on Images

- “My AI Wardrobe”
- “My Shave Assistant”
- Recognize money
- Train money literacy



BOT Services

- Chatbots
 - Q&A
 - Knowledge sharing
 - On Demand services
 - Without interrupting others
 - Confidence
- “My Artificial Mom”
 - Give confirmations
 - Reduce anxiety
 - Give structure



Language Understanding

- Analyze language
 - Broader picture
 - Custom words
 - Wrong/ misspelled/ mispronunciation
- Learn about the language of people
 - Understand what people mean
 - Find difficulty triggers



Speech

- Text-To-Speech
 - Custom Voice
 - Voice Assistants
- Read aloud for people
 - People with learning disabilities
 - Dyslexia
 - Give confirmations for anxiety



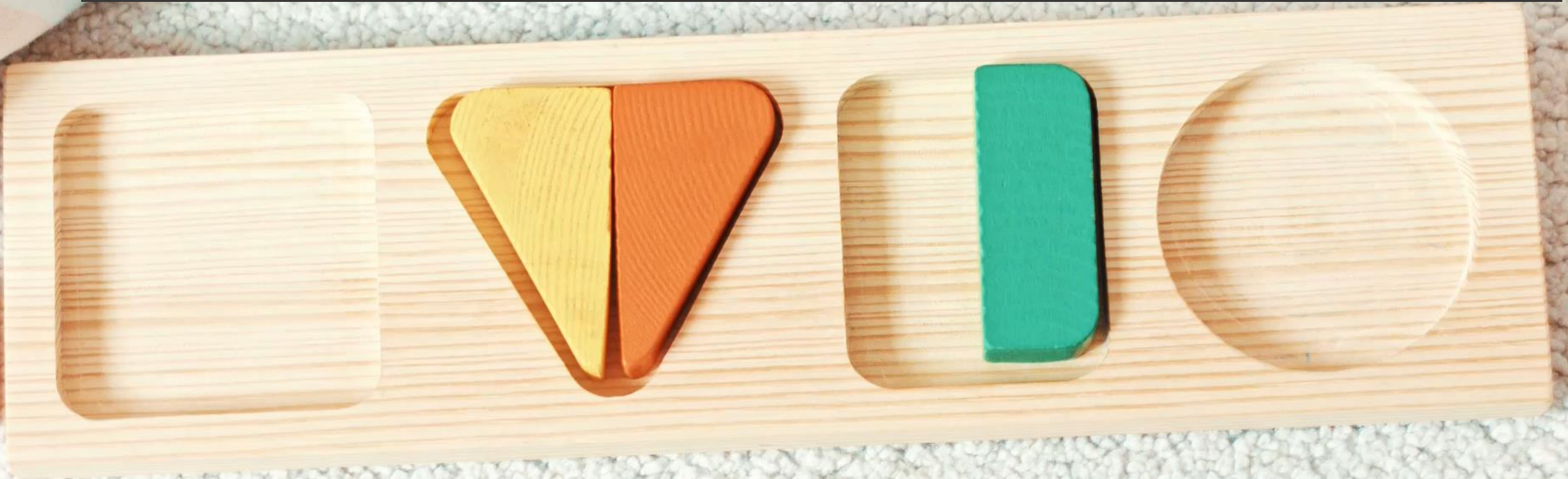
Translator

- Translating
- Reduce cognitive workload
- Present information
- Elderly

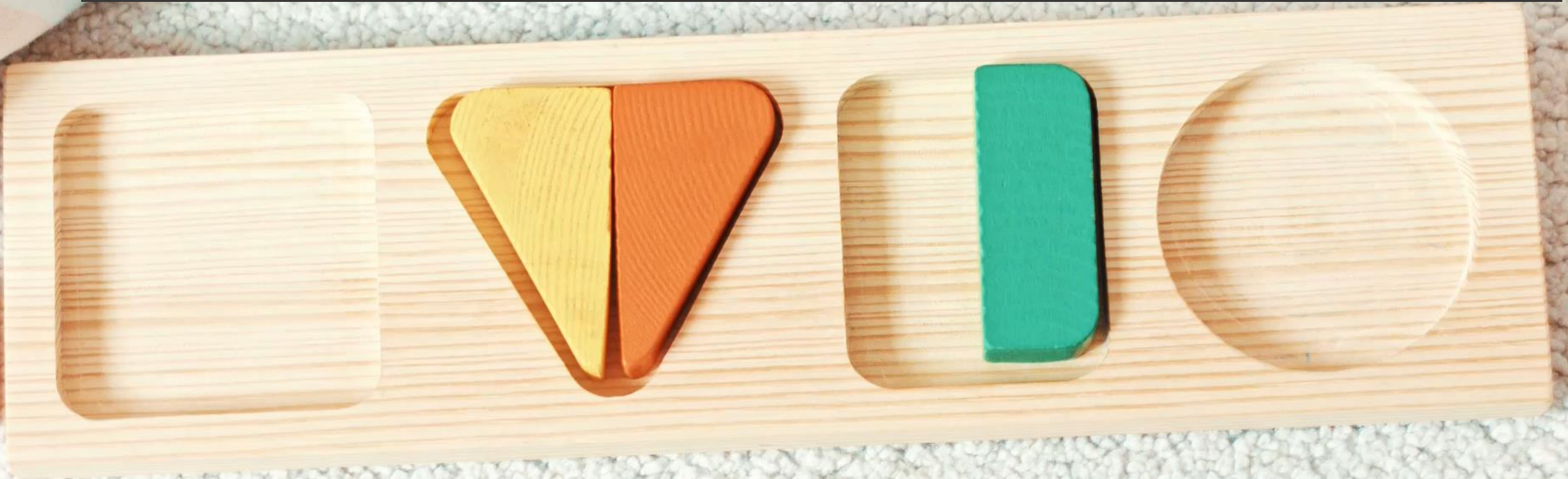
DEMO: Exploring Cognitive & Applied AI Services



DEMO: Creating a Speech Service



DEMO: Accessing Speech & Translate (SDK/ REST) in C#



Unsolved Real Life Problems to Fix

Literacy

- Difficulties with reading
- How do we translate governmental information?
- How do we provide information from shops/ leisure?
- Can we help with accessible communication?





Money

- 
- How do we make digital money tangible?
 - The pro's vs the con's of cashless
 - Pro
 - Sustainability
 - Limits diseases & infection
 - Con
 - Not tangible
 - Difficult concepts to learn
 - What about gifts from friends/ grandparents
 - What about a blocked bank account?

Daily Living

- Can we assist people with making daily choices?
- Can we help with purchases in the grocery store?
- Can we make temperature & humidity tangible?



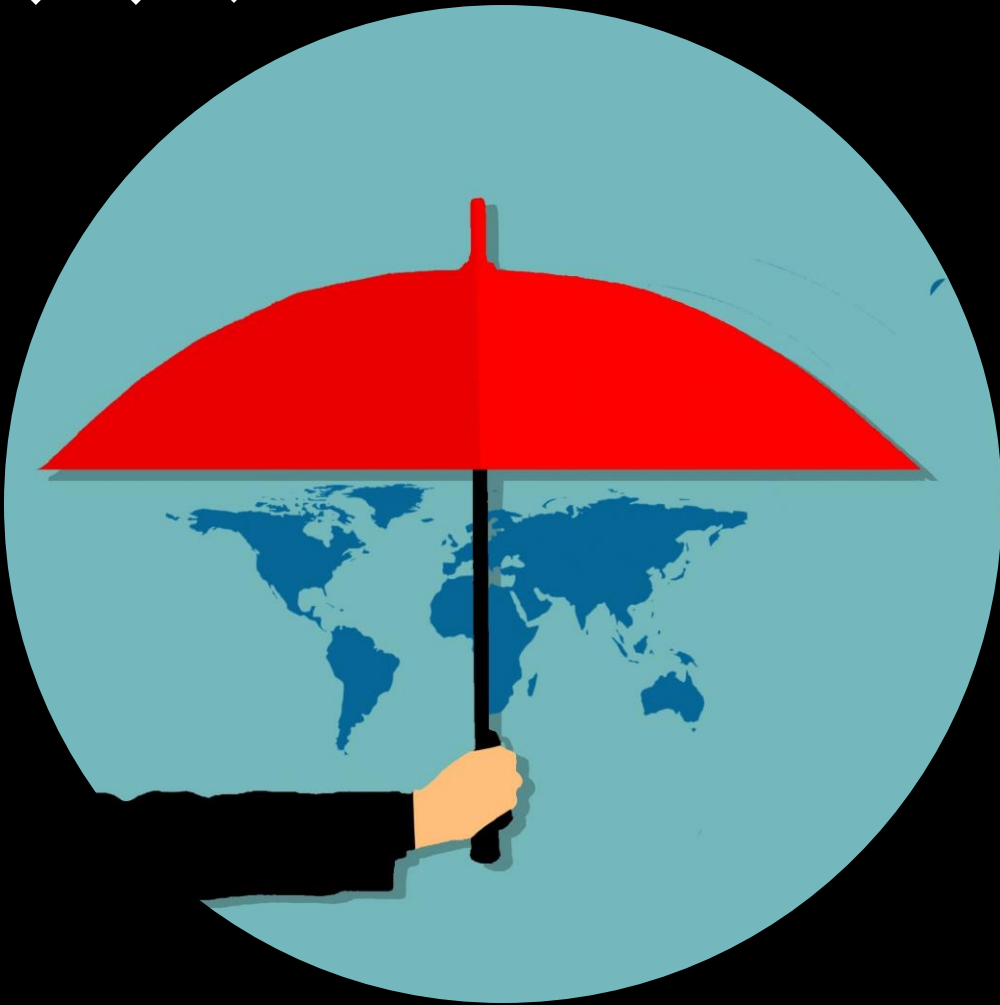
Feelings & Emotions

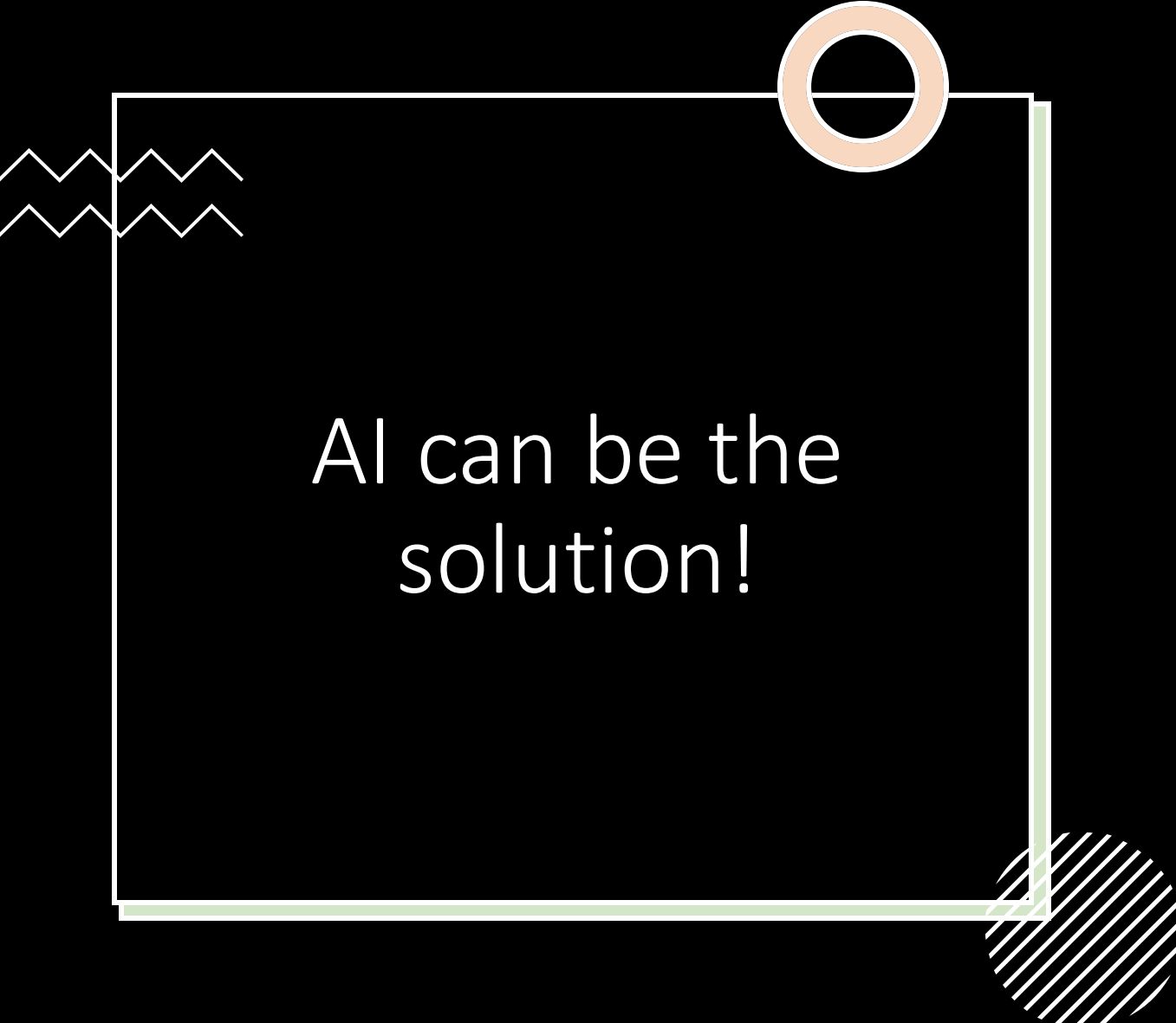
- Can we train emotions?
- Can we describe emotions?
- What about a 'negative mood' alarm?
- Can we pop-in happy news on bad days?



Dependency

- Can we limit external dependency?
- Can we help to train people themselves?
- What about over protecting parents?
- Are there taboos?





AI can be the
solution!

- Loads of solutions can be made with suggestions above
- Combined with existing AI tools (like Cognitive Services)
- It's just need to be build
- And the problems needs to be known!



What is done before?

Microsoft Seeing AI



Microsoft Translator



Office 365

People can add
images or adjust
content as needed



Inspiration for
your next
hack



The Immersive Recepie

Immersive Reader



My Vision Barber

- Custom Vision
- Face API



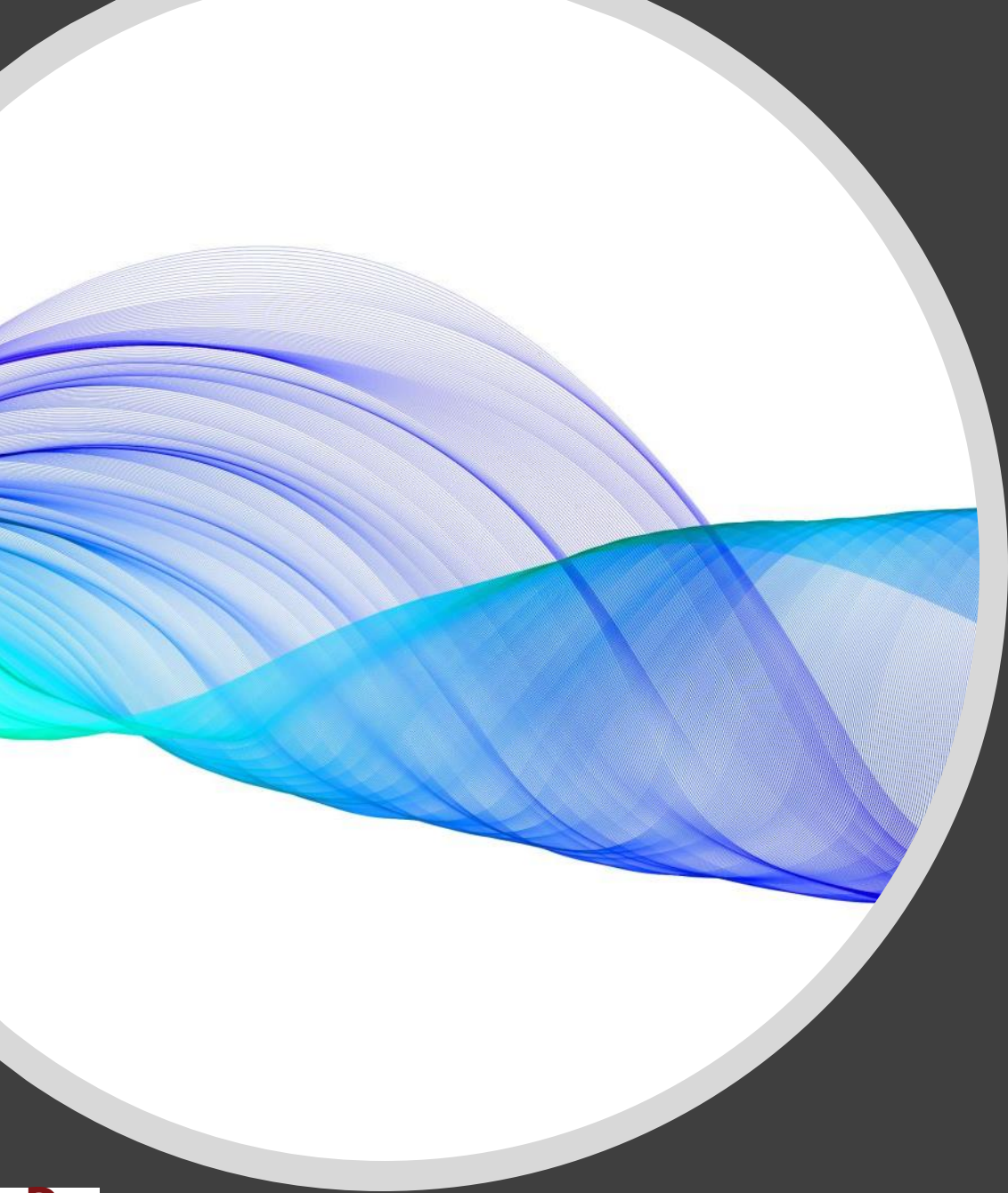
My Artificial Mom

- BOT Service
- Speech
- Custom Voice



My Smart Wardrobe

- Graph API
- Computer Vision
- Weather Forecasts



Recap



Thanks || Q&A

- @DennieDeclercq
- #DreamingIsBelieving
- dennie&@ddsoft.be

